INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, April - May, 2026

Degree: **B.C.A.** Semester: **Second**

Stream: **B.C.A.**

Paper Code: **BCACC202**

Paper Name: **Object Oriented Programming**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. Describe the different styles of writing prototypes. 2
2. How are data and functions organized in an object-oriented program? 2
3. What are the advantages of using the new operator as compared to the function malloc()? 2
4. What do you mean by dynamic binding? How is it useful in OOP? 2
5. Can we use the same function name for a member function of a class and an outside function in the same program file? If yes, how are they distinguished? If no, give reasons. 2
6. What is a class? How does it accomplish data hiding? 2
7. Describe the importance of destructors. 2
8. When is a friend function compulsory? Give an example. 2
9. Distinguish between the terms class template and template class. 2
10. When is a catch(...) handler used? 2
11. What is an exception? 2
12. List the applications of this pointer. 2

13. Express what is a stream. 2

14. Differentiate the operation of the following two statements: 2
cin >> c;
cin.get(c);

15. Illustrate the role of file() function; also explain the use of this function. 2

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Find errors, if any, in the following C++ statements: 5
(a) cout << "x=" x;
(b) m = 5; // n = 10; // s = m + n;
(c) cin >>x; >>y;
(d) cout << \n "Name:" << name;
(e) cout <<"Enter value:"; cin >> x;
(f) /*Addition*/ z = x + y;

17. Describe how data are shared by functions in a procedure-oriented program. 5

18. Write a program to read two numbers from the keyboard and display the larger value on the screen. 5

19. Develop an application in C++ to calculate the factorial of a given number n where the value of n is given from user end by implementing the concept of copy constructor. 5
Test case:
Enter the number to find factorial: 5
Printing from constructor:
Factorial of 5 is 120
Printing from copy constructor:
Factorial of 5 is 120

20. Write a program containing a possible exception. Use a try block to throw it and a catch block to handle it properly. 5

21. Write a program that demonstrates how certain exception types are not allowed to be thrown. 5

22. For a specific application, it is required to swap two data items. For different execution, it is required to swap variables of different types. But there is a 5

constraint to implement the above by a single function. Implement the above requirement using object-oriented programming.

Test case-1:
Input:
Enter two data to swap: 6
7

Output:
Before swap values are 6 and 7
After swap values are 7 and 6

Test case-2:
Input:
Enter two data to swap: 2.5
5.2

Output:
Before swap values are 2.5 and 5.2
After swap values are 5.2 and 2.5

23. Analyze that *cout* is able to display various types of data without any special instructions. **5**

24. Interpret the basic implementation difference between manipulators and ios member functions with a suitable example. **5**

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Construct a C++ program to store and display book information (title, ISBN, price, etc.) using an array of objects concept. **10**

26. An organization wants to calculate the total salary of two different types of employees: one is a manager and another is a sales person. Manager gets 60% DA and 30% HRA of basic salary. On the other hand, a sales person gets 40% DA and 20% HRA of basic salary. For both types of employees, 12% P. Tax will be deducted from their total salary. To solve this problem we must implement the concept of a friend function. **10**

Test Case:

Input:
Basic of Manager: Rs. 25000/-
Basic of Sales person: Rs. 20000/-

Output:
Total Salary of Manager: Rs. 41800/-
Total salary of Sales person: Rs. 28160/-
Total salary of both employees: Rs. 69960/-

27. Define a class to represent a bank account. Include the following members: **10**

Data members

a. Name of the depositor
b. Account number
c. Type of account
d. Balance amount in the account

Member functions

a. To assign initial values
b. To deposit an amount
c. To withdraw an amount after checking the balance
d. To display name and balance

Write a main program to test the program.

28. Write down a C++ program to store and display the information of all books (title, price etc.) of a book shop by implementation the concept of object and class where the total number of books will be given by book shopper as an input. **10**

Test Case:
Input:
The total number of books to store the information: 3
Enter details of book 1
Title: Programming in C
Price: 500
Enter details of book 2
Title: Programming in java
Price: 200
Enter details of book 3
Title: Computer Network
Price: 450

Output:

Details of all books
Book 1
Title: Programming in C
Price: 500
Book 2
Title: Programming in java
Price: 200
Book 3
Title: Computer Network
Price: 450

29. In a specific scenario, an application can have exceptions in three situations. For a specific variable, if the value is zero, the application needs to throw an integer exception; if the value is 1, then it will generate a float exception; and finally, if the value is -1, then it will generate a character exception. But while developing this application, it is to be remembered that there will be only one exception checking module. **10**

30. A mathematical research institute needs to implement a stack. But the requirement is such that for different times (executions) they need a stack of different types. For example, when integer type is selected the stack and item should be of type int. Same is applicable for float and character type. Implement the above requirement using object-oriented programming. **10**

31. Create a base class called shape. Use this class to store two *double* type values that could be used to compute the area of figures. Derive two specific classes called triangle and rectangle from the base shape. Add to the base class a member function *get_data()* to initialize base class data members and another member function *display_area()* to compute and display the area of figures. Make *display_area()* as a virtual function and redefine this function in the derived classes to suit their requirements.
Using these three classes, design a program that will accept dimensions of a triangle or a rectangle interactively, and display the area. **10**

32. Create a base class called employee to store the name and basic salary of an employee. Derive two classes called manager and engineer from the base class employee. Add to the base class a member function get_data() to initialize the employee details and another member function display_salary() to calculate and display the total salary. Make display_salary() a virtual function and redefine this function in the derived classes according to their salary structure. **10**

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